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Synergistic data compression

Abstract

A flush set of data pages to flush from a cache to non-volatile data storage is identified, and a compression group is identified that is made up of similar data pages within the flush set. A previously compressed data page is identified in non-volatile data storage that is similar to the data pages in the compression group. Data compression is performed on the compression group using a compression dictionary for the previously compressed data page. A flush set specific hash index and a global hash index may be used to identify the similar data pages in the compression group and the similar previously compressed data page. An indication of the similar previously compressed data page may be stored in the compressed block generated for the compression group, and the compression dictionary for the previously compressed data page need not be duplicated in the non-volatile data storage.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to technology for data reduction in a data storage system, and more specifically to performing data compression when flushing a set of data pages from cache to non-volatile data storage.

BACKGROUND

(2) Data storage systems are arrangements of hardware and software that are coupled to non-volatile data storage drives, such as solid state drives and/or magnetic disk drives. The data storage system services host I/O requests received from physical and/or virtual host machines (“hosts”). The host I/O requests received by the data storage system specify host data that is written and/or read by the hosts. The data storage system executes software that processes the host I/O requests by performing various data processing tasks to efficiently organize and persistently store the host data in the non-volatile data storage drives of the data storage system.

(3) Data storage systems use data compression to efficiently utilize their non-volatile data storage resources. Lossless data compression technology may be used to reduce the size of a set of user data by identifying and eliminating redundancy within the data. Examples of lossless data compression technologies include Lempel-Ziv (LZ) compression methods, such as LZ77, LZ78, DEFLATE, gzip, zstandard, Lempel-Ziv-Welch (LZW), etc. LZ and similar data compression technologies are examples of dictionary-based data compression, and operate by generating a compression dictionary of repeated character sequences found in the data being compressed. Data compression using such technologies includes generating an encoding of the data being compressed

by substituting compression dictionary entry identifiers (“codes”) for instances of the character sequences contained in the compression dictionary that are found in the data.

SUMMARY

(4) In the disclosed technology, a flush set of data pages to flush from a cache to non-volatile data storage is identified from within the cache. A compression group is then identified that is made up of data pages within the flush set that are similar. The disclosed technology also identifies a previously compressed data page that is stored in the non-volatile data storage and is similar to the data pages in the compression group. Data compression is performed on the compression group at least in part using a compression dictionary for the previously compressed data page.

(5) In some embodiments, identifying the compression group includes, for each individual data page in the flush set, applying a similarity hash to the data page to obtain a similarity hash value corresponding to the data page. The compression group is then identified as a set of data pages within the flush set that each have the same corresponding similarity hash value.

(6) In some embodiments, identifying the compression group includes generating a flush set hash index for the flush set. The flush set hash index is a searchable data structure that associates individual similarity hash values with corresponding data pages in the flush set. In such embodiments, the compression group is identified using the flush set hash index.

(7) In some embodiments, identifying the previously compressed data page that is stored in the non-volatile storage and is similar to the data pages in the compression group includes identifying a previously compressed data page that is stored in the non-volatile storage and has the same corresponding similarity hash value as the data pages in the compression group.

(8) In some embodiments, a global hash index is maintained for previously compressed data pages. The global hash index is a searchable data structure that associates individual similarity hash values with corresponding previously compressed data pages that are stored in the non-volatile data storage. In such embodiments, identifying the previously compressed data page that is similar to the data pages in the compression group is performed using the global hash index.

(9) In some embodiments, the disclosed technology performs data compression on the compression group using the compression dictionary for the previously compressed data page at least in part by accessing the compression dictionary for the previously compressed data page once from the previously compressed data page in non-volatile data storage into memory to create a copy of the compression dictionary for the previously compressed data page in memory, and discarding the copy of the compression dictionary for the previously compressed data page from memory when data compression on the compression group has completed.

(10) In some embodiments, the disclosed technology performs data compression on the compression group at least in part by generating a single compressed block that includes an indication of the previously compressed data page and an encoding of the data pages in the compression group, and storing the compressed block into the non-volatile data storage.

(11) In some embodiments, generating the compressed block further includes generating an additional dictionary component to compress portions of data in the data pages of the compression group that are not similar to the previously compressed data page, storing the additional dictionary component in the compressed block.

(12) In some embodiments, decompressing at least a portion of the compressed block to retrieve at least one of the data pages in the compression group includes accessing the compressed block in the non-volatile data storage, accessing the compression dictionary for the previously compressed data page based on the indication of the previously compressed data page stored in the compressed block, and decompressing at least a portion of the compressed block using the compression dictionary for the previously compressed data page.

(13) The disclosed technology is integral to providing a practical technical solution to the problem of efficiently providing high levels of data compression in a data storage system. By identifying a compression group of similar data pages within a flush set, and then compressing the group

together into a single compressed block, high compression ratios can be obtained. In addition, by utilizing the previously generated compression dictionary of a previously compressed data page that is similar to the data pages in the compression group during the compression of the compression group, the disclosed technology uses the synergy between the similar previously compressed data page and the similar data pages in the compression group to reduce the data storage system resources required to compress the compression group. Advantageously, the previously generated compression dictionary need not be stored again when the compressed block generated for the compression group is stored, as it remains persistently stored in association with (e.g. within) the similar previously compressed data page. Further improvement may be obtained during data retrieval with regard to the data pages in the compression group, since the compression dictionary for the previously compressed data page may be accessed again for data decompression using the indication of the previously compressed data page stored in the compressed block for the compression group.

(14) The foregoing summary does not indicate required elements, or otherwise limit the embodiments of the disclosed technology described herein. The technical features described herein can be combined in any specific manner, and all combinations may be used to embody the disclosed technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The objects, features and advantages of the disclosed technology will be apparent from the following description of embodiments, as illustrated in the accompanying drawings in which like reference numbers refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed on illustrating the principles of the disclosed technology.

(2) FIG. 1 is a block diagram showing an illustrative embodiment of the disclosed technology;

(3) FIG. 2 is a block diagram showing an example of the uncompressed data stored in a compression group and a similar previously compressed data page in an example of operation of the disclosed technology;

(4) FIG. 3 is a block diagram showing an example of a compressed block generated by the disclosed technology; and

(5) FIG. 4 is a flow chart showing an example of steps performed in some embodiments.

DETAILED DESCRIPTION

(6) Embodiments will now be described with reference to the figures. The embodiments described herein are not limiting, and are provided only as examples, in order to illustrate various features and principles of the disclosed technology. The embodiments of disclosed technology described herein are integrated into a practical solution to the problem of efficiently obtaining high levels of data compression in a data storage system.

(7) The disclosed technology operates by identifying a flush set of data pages to flush from a cache to non-volatile data storage, and then identifying a compression group within the flush set that is made up of similar data pages. A similar previously compressed data page is also identified. The previously compressed data page is stored in the non-volatile data storage and is similar to the data pages in the compression group. Data compression is performed on the compression group using a compression dictionary that was previously generated for the similar previously compressed data page.

(8) The compression group may be identified by, for each individual data page in the flush set, applying a similarity hash to that data page to obtain a corresponding similarity hash value. The compression group may be identified as a set of multiple data pages within the flush set that each

have the same corresponding similarity hash value.

(9) The compression group may be identified by generating a flush set hash index for the flush set. The flush set hash index is a searchable data structure that associates individual similarity hash values with corresponding data pages in the flush set. The compression group may be identified using the flush set hash index generated for the flush set.

(10) The previously compressed data page that is stored in the non-volatile storage and is similar to the data pages in the compression group may be identified by identifying a previously compressed data page that is stored in the non-volatile storage and has the same corresponding similarity hash value as the data pages in the compression group. To this end, a global hash index may be maintained for previously compressed data pages. The global hash index is a searchable data structure that associates individual similarity hash values with corresponding previously compressed data pages that are stored in the non-volatile data storage. Accordingly, identification of the previously compressed data page that is similar to the data pages in the compression group may be performed using the global hash index, and without having to decompress the previously compressed data page.

(11) Data compression may be performed on the compression group using the compression dictionary generated for the previously compressed data page at least in part by accessing the compression dictionary for the previously compressed data page once, e.g. reading the compression dictionary generated for the previously compressed data page into memory from within the previously compressed data page stored in non-volatile data storage, in order to create a memory copy of the compression dictionary generated for the previously compressed data page. The copy of the compression dictionary generated for the previously compressed data page from memory is then discarded when data compression on the compression group has completed.

(12) Data compression on the compression group may be performed by generating a single compressed block that includes an indication of the previously compressed data page and an encoding of all the data pages in the compression group. The compressed block is then stored in the non-volatile data storage.

(13) The compressed block may further be generated by generating an additional dictionary component to compress portions of data in the data pages of the compression group that are not similar to the previously compressed data page, and storing the additional dictionary component within the compressed block.

(14) Decompressing at least a portion of the compressed block to retrieve the uncompressed version of at least one of the data pages in the compression group may include accessing the compressed block in the non-volatile data storage, accessing the compression dictionary for the previously compressed data page based on the indication of the previously compressed data page stored in the compressed block, and decompressing at least a portion of the compressed block using the compression dictionary for the previously compressed data page.

(15) FIG. 1 shows an example of an operational environment in which embodiments of the disclosed technology may operate, and an illustrative embodiment of the disclosed technology. As shown in FIG. 1, one or more host computers (“Hosts”), shown as host computers **110(1)** through **110(N)**, and/or host applications executing in whole or in part on host computers **110(1)** through **110(N)**, access non-volatile data storage provided by Data Storage System **116**, e.g. over Network **114**. Data Storage System **116** includes at least one Storage Processor **120** and Non-Volatile Data Storage **118**. Data Storage System **116** may include one or more storage processors like Storage Processor **120**. Storage Processor **120** may be any type of physical or virtual computing device that is capable of processing host I/O requests.

(16) Non-Volatile Data Storage **118** may include or consist of solid state drives (SSDs), and/or some other specific type of storage drives, such as magnetic disk drives, electronic flash drives, optical drives, and/or other specific types of non-volatile data storage drives or devices.

(17) Network **114** may include or consist of any specific type of computer/communication network

and/or combination of networks.

(18) Hosts **110(1-N)** and/or host applications executing in whole or in part on Hosts **110(1-N)** logically connect to and communicate with the Storage Processor **120**. For example, Host I/O Requests **112** conveyed from Hosts **110(1-N)** to Storage Processor **120** may include block I/O requests and/or file I/O requests. Storage Processor **120** may be configured to receive host I/O requests through block-based and/or file-based storage protocols, and to respond to host I/O requests of either type by reading host data from and/or writing host data to the Non-Volatile Data Storage **118**.

(19) Storage Processor **120** includes Processing Circuitry **124**, Memory **130**, and communication interfaces. The communication interfaces may include adapters or the like that convert electronic and/or optical signals received over Network **114** into electronic form for use by Storage Processor **120**. Processing Circuitry **124** may be embodied as at least one processor core that is capable of independently reading and executing threads of program instructions as they are scheduled for execution. Processing Circuitry **124** may be integrated into a single central processing unit chip or chip package. Processing Circuitry **124** may be a subset of the processor cores contained in Storage Processor **120**, or may be the only processor core contained in Storage Processor **120**. Memory **130** may include both volatile memory (e.g., RAM), and/or non-volatile memory (e.g. ROM, disk drives, solid state drives, portions of Non-Volatile Storage **118**, etc.).

(20) Processing Circuitry **124** and Memory **130** together form specialized control circuitry, which is constructed and arranged to carry out the specific methods and functions described herein. As shown in FIG. **1**, Memory **130** stores software components and data structures that may be provided at least in part in the form of executable program instructions. When the executable instructions of the software components shown in FIG. **1** are executed by Processing Circuitry **124**, Processing Circuitry **124** performs the methods and functionality of the disclosed technology. Although certain specific software components are shown in FIG. **1**, those skilled in the art will recognize that Memory **130** may further include various other specific software components, which are not shown.

(21) In the example of FIG. **1**, for purposes of illustration, the software components in Memory **130** include Flush Logic **134**, which provides program logic for flushing data from Cache **132** into Non-Volatile Data Storage **118**.

(22) During operation of the illustrative embodiment shown in FIG. **1**, Flush Logic **134** identifies Flush Set of Data Pages **136** within Cache **132**, for the purpose of flushing Flush Set of Data Pages **136** from Cache **132** to Non-Volatile Data Storage **118**. The data pages in Flush Set of Data Pages **136** are dirty pages of host data in Cache **132** that need to be written to Non-Volatile Data Storage **118**, and may be identified based on various factors. For example, the data pages in each Flush Set of Data Pages **136** may be selected at least in part based on data access patterns of Hosts **110**, a duration of time the data pages have been stored in Cache **132**, a fullness level of Cache **132**, current availability of processing and/or other resources within Storage Processor **120**, and/or the logical proximity of the logical addresses of the data pages. Flush Set of Data Pages **136** may contain any specific number of data pages. Each data page may, for example, consist of 4 KB (four kilobytes) of host data. The disclosed technology is not limited in this regard, and any other specific page size may be used in the alternative.

(23) Flush Logic **134** performs data compression on some or all of the data pages in Flush Set of Data Blocks **136**, e.g. using Compression Logic **148**. In some embodiments, Flush Logic **134** also deduplicates the data pages in Flush Set of Data Blocks **136**, before using Compression Logic **148** to compress the data pages that remain in Flush Set of Data Blocks **136** after deduplication.

(24) Further during operation, Flush Logic **134** identifies Compression Group of Data Pages **138** within Flush Set of Data Pages **136**. Compression Group of Data Pages **138** is made up of some number of data pages within Flush Set of Data Pages **136** that Flush Logic **134** determines are similar to each other. Compression Group Of Data Pages **138** may, for example, consist of only a

portion (i.e. less than all) of the data pages in Flush Set of Data Pages **136**. Flush Logic **134** may identify the data pages contained in Compression Group of Data Pages **138** by, for each individual data page in Flush Set of Data Pages **136**, applying a similarity hash to the data page in order to obtain a similarity hash value corresponding to the data page. The similarity hash used may be any specific hash function that provides similarity hash values such that application of the hash function to sufficiently similar data pages has a high probability of returning the same similarity hash value. Accordingly, any data pages having the same similarity hash value are highly likely to be similar to each other. It is not necessary that data pages be completely identical in order for them to result in the same similarity hash value, only that the data pages be similar to each other to a predetermined degree. The specific amount of similarity between data pages that is required to result in the same similarity hash value may be an adjustable parameter. After applying the similarity hash function to each one of the data pages in Flush Set of Data Pages **136**, Flush Logic **134** identifies Compression Group of Data Pages **138** as a set of multiple data pages within Flush Set of Data Pages **136** that each have the same corresponding similarity hash value.

(25) In order to support efficient identification of Compression Group of Data Pages **138** within Flush Set of Data Pages **136**, Flush Logic **134** may also generate Flush Set Hash Index **144** for the Flush Set of Data Pages **136**. Flush Set Hash Index **144** is a searchable data structure that associates individual similarity hash values with corresponding data pages in Flush Set of Data Pages **136**. Compression Group of Data Pages **138** can then be identified using Flush Set Hash Index **144**, e.g. by finding a set of data pages in Flush Set of Data Pages **136** that all have the same corresponding similarity hash value.

(26) Flush Logic **134** also identifies a previously compressed data page that is i) stored in Non-Volatile Data Storage **118**, and ii) is similar to the data pages in Compression Group of Data Pages **138**. For example, Flush Logic **134** identifies Previously Compressed Data **138**. For example, Flush Logic **134** may identify Previously Compressed Data Page **140** by determining that Previously Compressed Data Page **140** has the same corresponding similarity hash value as the data pages in the compression group. To support identification of a similar previously compressed data page, Flush Logic **134** may maintain a global hash index for previously compressed data pages, shown by Global Hash Index **146**. Global Hash Index **146** is a searchable data structure that associates individual similarity hash values with corresponding previously compressed data pages that are stored in Non-Volatile Data Storage **118**. Flush Logic **134** may identify Previously Compressed Data Page **140** as being similar to the data pages in Compression Group of Data Pages **138** by searching the Global Hash Index **146** using the similarity hash value obtained for each of the data pages in Compression Group of Data Pages **138**. Such a search returns an identifier for any previously compressed data page that is similar to the data pages in Compression Group of Data Pages **140**, e.g. an identifier of Previously Compressed Data Page **140**.

(27) Flush Logic **134** then performs a single data compression operation for all of the data pages contained in Compression Group of Data Pages **138**, at least in part using a compression dictionary that was previously generated for Previously Compressed Data Page **140**. For example, the data compression operation performed on the Compression Group of Data Pages **138** may be performed using Compression Logic **148**. Any specific type of dictionary-based, lossless data compression may be used. In order to perform the compression operation on Compression Group of Data Pages **138** using the compression dictionary that was previously generated for Previously Compressed Data Page **140**, Flush Logic **134** may read the compression dictionary for Previously Compressed Data Page **140** from within the compressed block storing Previously Compressed Data Page **140** in Non-Volatile Data Storage into Memory **130** to create a copy of the compression dictionary for Previously Compressed Data Page **140**, as shown by Compression Dictionary for Previously Compressed Data Page **142**. Flush Logic **142** then passes Compression Dictionary for Previously Compressed Data Page **142** together with Compression Group of Data Pages **138** to Compression Logic **148** in order for Compression Logic **148** to compress Compression Group of Data Pages **138**

at least in part using Compression Dictionary for Previously Compressed Data Page **142**. When compression of Compression Group of Data Pages **138** has completed, Flush Logic **134** discards Compression Dictionary for Previously Compressed Data Page **142** from Memory **130**, in order to free up memory space.

(28) As further shown in FIG. **1**, Flush Logic **134** uses Compression Logic **148** to compress the data pages in Compression Group of Data Pages **138** into a single compressed block, shown by Compressed Block **150**. Compressed Block **150** includes an indication (e.g. identifier, name, address, and/or offset or the like) of Previously Compressed Data Page **140**, e.g. an indication of the location of the compressed block storing Previously Compressed Data Page **140** in Non-Volatile Data Storage **118**. Compressed Block **150** further includes an encoding of the data pages in Compression Group of Data Pages **138** that was generated by Compression Logic **148** at least in part using Compression Dictionary for Previously Compressed Data Page **142**. Flush Logic **134** then stores Compressed Block **150** into Non-Volatile Data Storage **118**, as indicated by reference number **152**.

(29) Generation of Compressed Block **150** by Flush Logic **134** using Compression Logic **148** may further include generating an additional dictionary component to compress portions of the data in the data pages in Compression Group of Data Pages **138** that are not similar to the data in Previously Compressed Data Page **140**. The additional dictionary component may also be stored in Compressed Block **150**. During compression of Compression Group of Data Pages **138**, encodings of the portions of data in Compression Group of Data Pages **138** that are dissimilar to Previously Compressed Data Page **140** may be generated based on the additional dictionary component, and are also stored in Compressed Block **150**.

(30) Subsequent decompression of at least a portion of Compressed Block **150** to retrieve at least one of the data pages in Compression Group of Data Pages **138** may include accessing Compressed Block **150** in Non-Volatile Data Storage **118**, accessing the compression dictionary generated for Previously Compressed Data Page **140** based on the indication of Previously Compressed Data Page **140** that was stored in Compressed Block **150**, and decompressing at least a portion of Compressed Block **150** using the compression dictionary for Previously Compressed Data Page **140**. In some embodiments, when only an individual data page of Compression Group of Data Pages **138** is to be retrieved, e.g. in response to a host read request, only the portion of Compressed Block **150** that contains the compressed data (i.e. encodings) for that individual data page is decompressed at least in part using the compression dictionary for Previously Compressed Data Page **140**, without decompression of the other compressed data that was generated for the other data pages in Compression Group of Data Pages **138** and stored within the Compressed Block **150**.

(31) FIG. **2** is a block diagram showing an example of data stored in a compression group of data pages (“Compression Group **200**”) and within a previously compressed data page (“Page D **202**”) that is determined to be similar to the data pages in Compression Group **200** during an operational example. While Page D **202** has previously been compressed at the time it is determined to be similar to the data pages in Compression Group **200**, the example of FIG. **2** illustrates and refers to the uncompressed version of Page D **202**, in order to explain the operation of the disclosed technology. In the example of FIG. **2**, the Compression Group **200** includes three uncompressed data pages, i.e. Page A **204**, Page B **206**, and Page C **208**. The contents of each uncompressed data page shown in FIG. **2** is described in terms of four 1 KB sections within that data page. For example, Page A **204** is made up of the four 1 KB sections **204a**, **204b**, **204c**, and **204d**, Page B **206** is made up of the four 1 KB sections **206a**, **206b**, **206c**, and **206d**, and Page C **208** is made up of the four 1 KB sections **208a**, **208b**, **208c**, and **208d**, and Page D **202** is made up of the four 1 KB sections **202a**, **202b**, **202c**, and **202d**.

(32) In the example of FIG. **2**, sections **202a** and **202b** in Page D **202** contain data “D”, which is the same as or similar to the data “D” contained in sections **204b** and **204c** of Page A **204**, sections **206c** and **206d** of Page B **206**, and sections **208b** and **208d** of Page C **208**. Sections **202c** and **202d**

of Page D **202** contains data that is unique to Page D **202**, e.g. the data “X”.

(33) Section **204a** in Page A **204** contains data “A”, which is the same as or similar to the data “A” contained in section **208c** of Page C **208**. Section **204d** in Page A **204** contains data “B”, which is the same as or similar to the data “B” contained in section **206a** of Page B **206**.

(34) Section **206b** in Page B **206** contains data “C”, which is the same as or similar to the data “C” contained in section **208a** of Page C **208**.

(35) The total uncompressed size of Page A **204** (4 KB), Page B **206** (4 KB), Page C **208** (4 KB), and Page D **202** (4 KB) is 16 KB. Using the technology disclosed herein, during compression of the Compression Group **200**, the 2 KB of data “D” in sections **204b** and **204c** of Page A **204**, sections **206c** and **206d** of Page B **206**, and sections **208b** and **208d** of Page C **208**, that is the same as or similar to the 2 KB of data “D” in Page D **202**, is compressed using the compression dictionary that was previously generated and used when Page D **202** was previously compressed. With an 80% data reduction, the 2 KB of data “D” in each one of Page A **204**, Page B **206**, Page C **208**, and Page D **202** (8 KB total) is reduced to 0.4 KB of compressed data for each page one of Page A **204**, Page B **206**, Page C **208**, and Page D **202** (1.6 KB total).

(36) The 2 KB of data “A” in section **204a** of Page A **204** and section **208c** of Page C **208**, the 2 KB of data “B” in section **204d** in Page A **204** and section **206a** in Page B **206**, and the 2 KB of data “C” in section **206b** in Page B **206** and section **208a** in Page C **208**, are not the same as or similar to the data in Page D **202**, and accordingly are not compressed using the compression dictionary that was previously generated when Page D **202** was previously compressed. Instead, those sections of Page A **204**, Page B **206**, and Page C **208** are compressed using an additional dictionary component that is generated during the compression of Compression Group **200**, based on data “A”, “B”, and “C”. With an 80% data reduction, the 6 KB of uncompressed data “A”, “B”, and “C” in Page A **204**, Page B **206**, and Page C **208** is reduced to 0.4 KB of compressed data for each page (1.2 KB total).

(37) Accordingly, in a system that includes the disclosed technology, the size of the compressed data for Page A **204**, Page B **206**, Page C **208**, and Page D **202** is: 1.6 KB (compressed data “D”)+1.2 KB (compressed data “A”, “B”, and “C”)+2 KB (unique data “X” in Page D **202**)=4.8 KB for a data reduction of 16 KB-4.8 KB=11.2 KB.

(38) FIG. 3 is a block diagram showing an example of a Compressed Block **300** generated by the disclosed technology to compress a corresponding compression group of data pages during a flush operation. The Compressed Block **300** is an example of the Compressed Block **150** in FIG. 1.

(39) As shown in the example of FIG. 3, Compressed Block **300** includes a Header **302**, Metadata **304**, Additional Dictionary Component **306**, and Tail **308**. Header **302** may include an indication of the previously compressed data page that was identified as similar to the corresponding compression group for which Compressed Block **300** was generated, e.g. an identifier, name, address, and/or offset or the like that enables the compression dictionary for the similar previously compressed data page to be retrieved when one or more of the data pages in the corresponding compression group needs to be retrieved and decompressed. Alternatively, such an indication may be included in some other part of Compressed Block **300**. Header **302** may also include system specific information, such an indication of the specific compression algorithm used to compress the corresponding compression group, and/or other system specific information.

(40) Metadata **304** includes an encoding of the data in the data pages of the corresponding compression group, e.g. an encoding of the data in the data pages of the corresponding compression group generated based on mappings between compression dictionary entry identifiers and data sequences contained in the compression dictionary for the similar previously compressed data page, and/or based on mappings between compression dictionary entry identifiers and data sequences contained in the Additional Dictionary Component **306**.

(41) Additional Dictionary Component **306** contains mappings between dictionary entries and data sequences generated when portions of the data in the data pages of the corresponding compression

group were compressed that were not similar to the data in the previously compressed data page.

(42) Tail **308** includes additional system specific information, e.g. data verification information such as one or more checksums or the like for the contents of Compressed Block **300**.

(43) FIG. **4** is a flow chart showing an example of steps performed in some embodiments.

(44) At step **400**, a flush set of data pages is identified to flush from a cache to non-volatile data storage.

(45) At step **402**, a compression group is identified that is made up of data pages within the flush set that are similar to each other.

(46) At step **404**, a previously compressed data page is identified that is stored in the non-volatile data storage that is similar to the data pages in the compression group.

(47) At step **406**, a single data compression operation is performed on all the data pages in the compression group at least in part using a compression dictionary generated for the previously compressed data page.

(48) At step **408**, a compressed block generated at step **406** is stored into the non-volatile data storage.

(49) At step **410**, the compressed block is accessed in non-volatile data storage and at least a portion of the compressed block is decompressed using the compression dictionary for the previously compressed data page to retrieve one or more of the data pages in the compression block.

(50) As will be appreciated by those skilled in the art, aspects of the technology disclosed herein may be embodied as a system, method, or computer program product. Accordingly, each specific aspect of the present disclosure may be embodied using hardware, software (including firmware, resident software, micro-code, etc.) or a combination of software and hardware. Furthermore, aspects of the technologies disclosed herein may take the form of a computer program product embodied in one or more non-transitory computer readable storage medium(s) having computer readable program code stored thereon for causing a processor and/or computer system to carry out those aspects of the present disclosure.

(51) Any combination of one or more computer readable storage medium(s) may be utilized. The computer readable storage medium may be, for example, but not limited to, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any non-transitory tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(52) The figures include block diagram and flowchart illustrations of methods, apparatus(s) and computer program products according to one or more embodiments of the invention. It will be understood that each block in such figures, and combinations of these blocks, can be implemented by computer program instructions. These computer program instructions may be executed on processing circuitry to form specialized hardware. These computer program instructions may further be loaded onto programmable data processing apparatus to produce a machine, such that the instructions which execute on the programmable data processing apparatus create means for implementing the functions specified in the block or blocks. These computer program instructions may also be stored in a computer-readable memory that can direct a programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture including instruction means which implement the function specified in the block or blocks. The computer program instructions may also be loaded onto a programmable data processing apparatus to cause a series of operational steps to be performed on the programmable apparatus to produce a computer implemented process such that the instructions which execute on the programmable apparatus provide steps for implementing the

functions specified in the block or blocks.

(53) Those skilled in the art should also readily appreciate that programs defining the functions of the present invention can be delivered to a computer in many forms; including, but not limited to: (a) information permanently stored on non-writable storage media (e.g. read only memory devices within a computer such as ROM or CD-ROM disks readable by a computer I/O attachment); or (b) information alterably stored on writable storage media (e.g. floppy disks and hard drives).

(54) While the invention is described through the above exemplary embodiments, it will be understood by those of ordinary skill in the art that modification to and variation of the illustrated embodiments may be made without departing from the inventive concepts herein disclosed.

Claims

1. A method comprising: identifying a flush set of data pages to flush from a cache to non-volatile data storage; identifying a compression group made up of data pages within the flush set that are similar; identifying a previously compressed data page that is stored in the non-volatile data storage that is similar to the data pages in the compression group; and performing data compression on the compression group at least in part using a compression dictionary for the previously compressed data page.

2. The method of claim 1, wherein identifying the compression group further comprises: for each individual data page in the flush set, applying a similarity hash to the data page to obtain a similarity hash value corresponding to the data page; and identifying the compression group as a plurality of data pages within the flush set that have the same corresponding similarity hash value.

3. The method of claim 2, wherein identifying the compression group further comprises: generating a flush set hash index for the flush set, wherein the flush set hash index is a searchable data structure that associates individual similarity hash values with corresponding data pages in the flush set; and identifying the compression group using the flush set hash index.

4. The method of claim 3, wherein identifying the previously compressed data page that is stored in the non-volatile storage and is similar to the data pages in the compression group comprises identifying a previously compressed data page that is stored in the non-volatile storage and has the same corresponding similarity hash value as the data pages in the compression group.

5. The method of claim 4, further comprising: maintaining a global hash index for previously compressed data pages, wherein the global hash index is a searchable data structure that associates individual similarity hash values with corresponding previously compressed data pages that are stored in the non-volatile data storage; and identifying the previously compressed data page that is similar to the data pages in the compression group using the global hash index.

6. The method of claim 5, wherein performing the data compression on the compression group using the compression dictionary for the previously compressed data page further comprises: accessing the compression dictionary for the previously compressed data page once from the previously compressed data page in non-volatile data storage into memory to create a copy of the compression dictionary for the previously compressed data page in memory; and discarding the copy of the compression dictionary for the previously compressed data page from memory when data compression on the compression group has completed.

7. The method of claim 6, further comprising: performing data compression on the compression group at least in part by generating a compressed block that includes an indication of the previously compressed data page and an encoding of the data pages in the compression group; and storing the compressed block into the non-volatile data storage.

8. The method of claim 7, wherein generating the compressed block further comprises: generating an additional dictionary component to compress portions of data in the data pages of the compression group that are not similar to the previously compressed data page; and storing the additional dictionary component in the compressed block.

9. The method of claim 8, further comprising decompressing at least a portion of the compressed block to retrieve at least one of the data pages in the compression group, wherein decompressing at least a portion of the compressed block comprises: accessing the compressed block in the non-volatile data storage; accessing the compression dictionary for the previously compressed data page based on the indication of the previously compressed data page stored in the compressed block; and decompressing at least a portion of the compressed block using the compression dictionary for the previously compressed data page.

10. A data storage system, comprising: processing circuitry and memory coupled to the processing circuitry, the memory storing instructions, wherein the instructions, when executed by the processing circuitry, cause the processing circuitry to: identify a flush set of data pages to flush from a cache to non-volatile data storage; identify a compression group made up of data pages within the flush set that are similar; identify a previously compressed data page that is stored in the non-volatile data storage that is similar to the data pages in the compression group; and perform data compression on the compression group at least in part using a compression dictionary for the previously compressed data page.

11. The data storage system of claim 10, wherein the program code, when executed by the processing circuitry, causes the processing circuitry to identify the compression group by causing the processing circuitry to: for each individual data page in the flush set, apply a similarity hash to the data page to obtain a similarity hash value corresponding to the data page; and identify the compression group as a plurality of data pages within the flush set that have the same corresponding similarity hash value.

12. The data storage system of claim 11, wherein the program code, when executed by the processing circuitry, further causes the processing circuitry to identify the compression group by causing the processing circuitry to: generate a flush set hash index for the flush set, wherein the flush set hash index is a searchable data structure that associates individual similarity hash values with corresponding data pages in the flush set; and identify the compression group using the flush set hash index.

13. The data storage system of claim 12, wherein the previously compressed data page that is stored in the non-volatile storage and is similar to the data pages in the compression group is identified by identifying a previously compressed data page that is stored in the non-volatile storage and has the same corresponding similarity hash value as the data pages in the compression group.

14. The data storage system of claim 13, wherein the program code, when executed by the processing circuitry, further causes the processing circuitry to: maintain a global hash index for previously compressed data pages, wherein the global hash index is a searchable data structure that associates individual similarity hash values with corresponding previously compressed data pages that are stored in the non-volatile data storage; and identify the previously compressed data page that is similar to the data pages in the compression group using the global hash index.

15. The data storage system of claim 14, wherein the program code, when executed by the processing circuitry, causes the processing circuitry to perform the data compression on the compression group using the compression dictionary for the previously compressed data page by causing the processing circuitry to: access the compression dictionary for the previously compressed data page once from the previously compressed data page in non-volatile data storage into memory to create a copy of the compression dictionary for the previously compressed data page in memory; and discard the copy of the compression dictionary for the previously compressed data page from memory when data compression on the compression group has completed.

16. The data storage system of claim 15, wherein the program code, when executed by the processing circuitry, further causes the processing circuitry to: perform data compression on the compression group at least in part by generating a compressed block that includes an indication of the previously compressed data page and an encoding of the data pages in the compression group;

and store the compressed block into the non-volatile data storage.

17. The data storage system of claim 16, wherein the program code, when executed by the processing circuitry, further causes the processing circuitry to generate the compressed block by causing the processing circuitry to: generate an additional dictionary component to compress portions of data in the data pages of the compression group that are not similar to the previously compressed data page; and store the additional dictionary component in the compressed block.

18. The data storage system of claim 17, wherein the program code, when executed by the processing circuitry, further causes the processing circuitry to decompress at least a portion of the compressed block to retrieve at least one of the data pages in the compression group, and wherein the program code, when executed by the processing circuitry, further causes the processing circuitry to decompress at least a portion of the compressed block by causing the processing circuitry to: access the compressed block in the non-volatile data storage; access the compression dictionary for the previously compressed data page based on the indication of the previously compressed data page stored in the compressed block; and decompress at least a portion of the compressed block using the compression dictionary for the previously compressed data page.

19. A computer program product, comprising: a non-transitory, computer-readable medium having instructions stored thereon that, when executed by processing circuitry in a data storage system, cause the processing circuitry to perform a method comprising the steps of: identifying a flush set of data pages to flush from a cache to non-volatile data storage; identifying a compression group made up of data pages within the flush set that are similar; identifying a previously compressed data page that is stored in the non-volatile data storage that is similar to the data pages in the compression group; and performing data compression on the compression group at least in part using a compression dictionary for the previously compressed data page.
